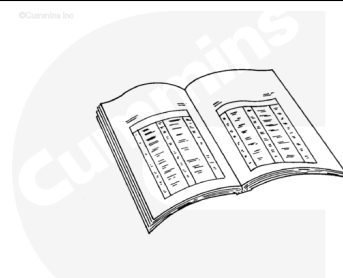


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ CAUTION ⚠

The two camshaft idler gears, camshaft gears, and crankshaft gear have index marks that are aligned during assembly. Improper alignment will cause extensive engine damage.

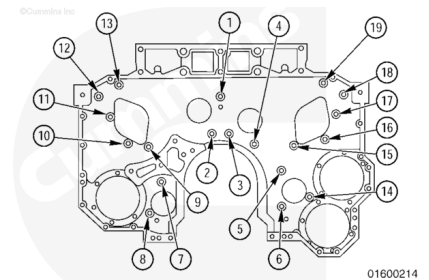
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>).
- Bar the engine over to align the index marks. Align the "V" on the idler gear with the "X" on the crankshaft gear. Align the "X" on the camshaft gear at the 12 o'clock position and with the dowel holes in the front gear housing.
- Remove the water pump. Refer to Procedure 008-062 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5 (</qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html>).
- Remove the left bank camshaft idler gear. Refer to Procedure 001-037 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-037-tr.html>).
- Remove the right bank camshaft idler gear. Refer to Procedure 001-038 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-038-tr.html>).

- Remove the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1
(/qs3/pubsys2/xml/en/procedures/56/56-001-039-tr.html).
- Remove the water pump idler gear. Refer to Procedure 001-040 in Section 1
(/qs3/pubsys2/xml/en/procedures/56/56-001-040-tr.html).
- Remove the left and right bank camshaft gears. Refer to Procedure 001-012 in Section 1
(/qs3/pubsys2/xml/en/procedures/56/56-001-012.html).

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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Remove the 19 front gear housing capscrews.

Remove the front gear housing and gasket.

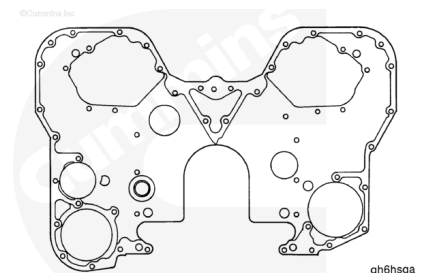
Discard the gasket.

Note : The dowel pins do not need to be removed unless they are damaged.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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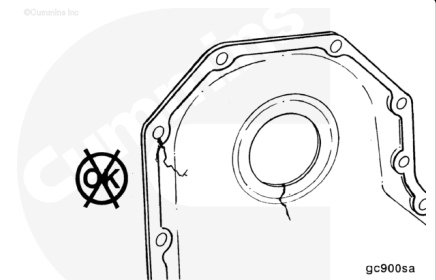
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the front gear housing with solvent and dry with compressed air.

Make sure the mating surfaces of the front gear housing do **not** have nicks or cracks.

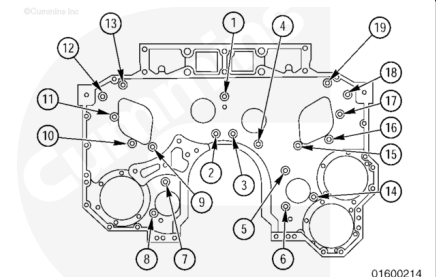
If the front gear housing is damaged, repair or replace the housing.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Locate the gear housing gasket and gear housing on the locating dowels.

Note : Loctite™ 242, Part Number 3824040, or equivalent, **must** be applied to the capscrews during installation. Loctite™ 242 is **not** required if the capscrews being used are new capscrews with a Loctite™ patch applied to the capscrew threads.

Install the mounting capscrews.

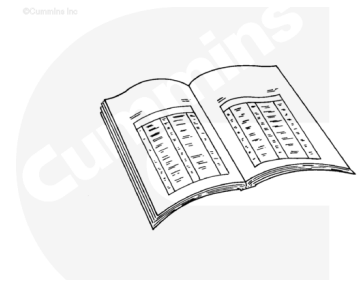
Use the sequence shown in the illustration and tighten the capscrews.

Torque Value: 80 n•m [59 ft-lb]

Finishing Steps

⚠ WARNING ⚠

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- Install the left and right bank camshaft gears. Refer to Procedure 001-012 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-012.html>).
- Install the water pump idler gear. Refer to Procedure 001-040 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-040-tr.html>).
- Install the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-039-tr.html>).
- Install the right bank idler gear. Refer to Procedure 001-038 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-038-tr.html>).
- Install the left bank idler gear. Refer to Procedure 001-037 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-037-tr.html>).
- Install the fuel pump. Refer to Procedure 005-016 in Section 5 (</qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html>).
- Install the water pump. Refer to Procedure 008-062 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Install the front gear cover. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>).

